Week 1 Report

Lab Setup, Linux and Networking Foundations

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1. Summary

In Week 1 I built a safe, isolated home lab in VirtualBox, practised core Linux administration, mapped the lab network, and worked through OverTheWire Bandit and eight TryHackMe rooms. I also wrote eight theory notes, a Linux command cheat sheet and two LinkedIn posts. All work used free tools only, and no Bandit passwords are published.

2. Deliverables at a glance

Area	What was completed
Lab setup	3 VMs (Kali, Ubuntu, Windows evaluation) on one isolated network, pings confirmed, a clean snapshot of each
Linux practice	3 users, 2 groups, shared folder with correct permissions, find / grep / ps / ss, cron job, 33+ commands documented
Network map	IP, gateway and DNS recorded for each VM and the lab diagram drawn in draw.io
OverTheWire Bandit	Levels 0 to 10 completed, one writeup and screenshots per level
TryHackMe	8 rooms completed, one writeup per room
Theory notes	8 notes, one file per question
Cheat sheet	Linux Command, What It Does, When to Use It (30+ commands)
LinkedIn	Concept post (SOC analyst vs penetration tester) and weekly post

3. Lab setup

I installed VirtualBox on a PC with 8 GB of RAM and created three virtual machines on one isolated network (192.168.98.0/24). I confirmed connectivity with ping between the VMs and took a snapshot of each clean VM.

VM	Memory	IP address	DNS
Ubuntu	2 GB	192.168.98.129	192.168.98.1
Kali Linux	2 GB	192.168.98.130	192.168.98.1
Windows 10	4 GB	192.168.98.131	192.168.98.1

Why isolated: vulnerable machines must never be bridged to a real network, because bridging exposes them to every device on that network and lets attack tools reach devices I do not own.

4. Linux practice (Ubuntu)

- Created three users (alice, john, carol) and two groups (developers, analysts). I used john instead of bob because bob was already the administrator account.
- Created the shared folder /srv/shared owned by root:developers with chmod 3770 (SGID, sticky bit, full access for owner and group, none for others).
- Result: users in developers could create files. carol, who is not in developers, received Permission denied. My admin account also needed sudo to list the folder.
- Used find, grep, ps and ss -tulpn, and scheduled a cron job.
- Documented 33+ commands with their output.

5. Network map

I recorded the IP, gateway and DNS of each VM using ip a, ip route, ping, traceroute, nslookup and dig (ipconfig and tracert on Windows) and drew the lab diagram in draw.io. On Ubuntu, ss -tulpn showed which services were listening:

Port	Service	Reachable from the network?
22 (TCP)	SSH	Yes
5353 (UDP)	mDNS	Yes
53 (TCP/UDP)	Local DNS resolver	No, localhost only
631 (TCP)	Printing service (CUPS)	No, localhost only
323 (UDP)	Time synchronisation	No, localhost only

6. OverTheWire Bandit and TryHackMe

- **Bandit levels 0 to 10:** completed over SSH with a screenshot and writeup for each level. Passwords are not published.
- **TryHackMe rooms:** Search Skills, Learning Cyber Security, Linux Fundamentals Part 1, What is Networking, Introductory Networking, Networking Concepts, Pyramid of Pain, Intro to LAN.

7. Theory notes

Note	Topic
01	What is cybersecurity and the CIA triad
02	SOC analyst vs VAPT / penetration tester vs incident responder
03	OSI vs TCP/IP model, with an attack per layer
04	IP, MAC, DNS, DHCP, ports and 12 common ports
05	Linux file system hierarchy and permissions (chmod, chown, SUID)
06	Threat vs vulnerability vs risk vs exploit
07	VM networking modes and why vulnerable labs are never bridged

Note	Topic
08	Legal and ethical basics, including Nigeria's Cybercrimes Act

8. Key Learnings

- Isolation keeps a lab safe: vulnerable VMs belong on Host-only or NAT Network, never bridged.
- Permissions are security. One wrong chmod can expose or lock a folder.
- Every open port is attack surface, so I check listening services with ss -tulpn.
- Small commands such as grep, sort and uniq become powerful when combined.
- Legal authorization and a defined scope are what separate ethical testing from a crime.

9. Compliance with the rules

- Free tools only: VirtualBox, Kali Linux, Ubuntu, Windows evaluation, draw.io, TryHackMe and OverTheWire.
- No Bandit passwords are published in any writeup.
- All notes and writeups are written in my own words.